UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 001 · Origin of electrode potential, cell potential · compiled 17 October 2026

Contents

1. Short notes
2. Keyword index
3. Rapid recall
4. Last 10 minutes

1 Short notes

1 · Electrochemistry and the two types of cell

Electrochemistry: interconversion of chemical and electrical energy. Galvanic cell: spontaneous redox reaction → electrical energy (batteries, fuel cells). Electrolytic cell: external energy drives a non-spontaneous reaction (electroplating). Needs: a redox reaction; anode (oxidation) and cathode (reduction); an ionic electrolyte (solution, melt or solid).

2 · Electrode potential and its origin: the Helmholtz double layer

Electrode potential E: the potential at the interface of a metal and a solution of its own ions. Origin: ionisation $\mathrm{M} \rightleftharpoons \mathrm{M^{n+}} + \mathrm{ne^-}$ and deposition $\mathrm{M^{n+}} + \mathrm{ne^-} \rightleftharpoons \mathrm{M}$ compete. Case I (ionisation faster): metal negative, attracts cations. Case II (deposition faster): metal positive, attracts anions. The two charge layers form the Helmholtz electrical double layer.

3 · Standard electrode potential, electrochemical cell and cell notation

E°: electrode potential at unit concentration, 298 K and 1 atm for gases; depends only on the electrode. Two electrodes form a cell (single potentials are not measurable). Notation: anode | anode solution || cathode solution | cathode, e.g. $\mathrm{Zn|Zn^{2+}||Cu^{2+}|Cu}$.

4 · Cell potential, standard cell potential and free energy

E_{cell} = difference of the electrode potentials = $E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$; standard (1 M, 1 atm, 298 K): E°_{cell}. $\Delta G = -nFE_{cell}$: spontaneous $\Leftrightarrow$ $\Delta G < 0 \Leftrightarrow E_{cell} > 0$. Daniell: $E^\circ_{cell} = 1.10$ V, $\Delta G^\circ = -212.3$ kJ mol^{-1}.

5 · The electrochemical series and its uses

Electrochemical series: electrodes in order of standard reduction potential, Li $\mathbf{-3.03}$ ... $\mathbf{H_2}$ 0.00 ... Au $\mathbf{+1.50}$ V (slide). Negative = oxidation tendency (strong reducing metals); positive = reduction tendency. Lower $\boldsymbol{E^\circ}$ = anode. Uses: $\boldsymbol{E^\circ_{cell}}$, spontaneity, displacement, corrosion (Zn protects Fe; Sn only while intact).

2 Keyword index

26 terms; the number in brackets is the topic.

Anode (1). Electrode where oxidation occurs; left in the notation; lower E°.

Cathode (1). Electrode where reduction occurs; right in the notation; higher E°.

Cell notation (3). anode | solution || solution | cathode.

Cell potential (4). Difference of the two electrode potentials, E_{cell}.

Daniell cell (3). $\mathrm{Zn|Zn^{2+}||Cu^{2+}|Cu}$; $E^\circ_{cell} = 1.10$ V.

Deposition (2). Reduction $\mathrm{M^{n+} + ne^- \rightarrow M}$ at the electrode.

Driving force (4). E_{cell}; related to $\Delta G = -nFE_{cell}$.

Electrochemical cell (3). Two electrodes coupled so that a potential difference can be measured.

Electrochemical series (5). Electrodes ordered by standard reduction potential.

Electrochemistry (1). Interconversion of chemical and electrical energy.

Electrode potential (2). Potential at the metal–solution interface, E.

Electrolyte (1). Conducts by movement of ions: solution, melt or solid.

Electrolytic cell (1). Electrical → chemical energy; non-spontaneous reaction.

Faraday constant (4). $F = 96500$ C mol^{-1}.

Galvanic cell (1). Chemical → electrical energy; spontaneous reaction.

Galvanising (5). Zn coating on Fe; Zn is sacrificial since $E^\circ_{Zn} < E^\circ_{Fe}$.

Helmholtz double layer (2). Charged metal + layer of opposite ions at the interface.

Ionisation (2). Oxidation $\mathrm{M \rightarrow M^{n+} + ne^-}$ at the electrode.

Phase boundary (3). Single line | in the notation.

Salt bridge (3). Double line || ; connects the two solutions.

SHE (5). Standard hydrogen electrode, $E^\circ = 0.00$ V (reference).

Solid electrolyte (1). e.g. $\mathrm{Y_2O_3}$-doped zirconia ($\mathrm{O^{2-}}$ conduction).

Spontaneous (4). $\Delta G < 0$, $E_{cell} > 0$.

Standard cell potential (4). E°_{cell} at 1 M, 1 atm, 298 K.

Standard electrode potential (3). E° at unit concentration, 298 K, 1 atm.

Tinning (5). Sn coating on Fe; cathodic, Fe corrodes at a scratch.

3 Rapid recall

1	Define electrochemistry.	Interconversion of chemical energy and electrical energy.
2	Galvanic cell converts ...?	Chemical energy to electrical energy (spontaneous reaction).
3	Electrolytic cell converts ...?	Electrical energy to chemical energy (non-spontaneous reaction).
4	Three requirements of an electrochemical system?	Redox reaction, electrodes, electrolyte (ionic conduction).
5	Define electrode potential.	Potential at the interface of a metal and a solution of its own ions.
6	Ionisation faster: charge on metal?	Negative; attracts positive ions.
7	Deposition faster: charge on metal?	Positive; attracts negative ions.
8	Name of the interfacial charge layer?	Helmholtz electrical double layer.
9	Standard conditions for E°?	Unit concentration, 298 K, 1 atm for gases.
10	Why couple two electrodes?	A single electrode potential cannot be measured.
11	Daniell cell notation?	$\mathrm{Zn}\vert\mathrm{Zn^{2+}(1\,M)}\vert\vert\mathrm{Cu^{2+}(1\,M)}\vert\mathrm{Cu}$.
12	E_{cell} formula?	$E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$.
13	E°_{cell} of the Daniell cell?	1.10 V.
14	ΔG and E_{cell}?	$\Delta G = -nFE_{cell}$.
15	ΔG° of the Daniell cell?	-212.3 kJ mol^{-1}.
16	Spontaneous reaction: sign of E_{cell}?	Positive ($\Delta G < 0$).
17	Negative E° indicates?	Oxidation tendency.
18	Top and bottom of the slide series?	Li -3.03 V; Au $+1.50$ V.
19	Fe/Au E°_{cell} with ESA data?	$1.52 + 0.44 = 1.96$ V ($n = 6$).
20	Galvanised iron scratched: does Fe corrode?	No; Zn (-0.76 V) corrodes sacrificially.
21	Tinned iron scratched?	Fe (-0.44 V) is the anode and corrodes rapidly.

Mnemonic (study aid). *"An Ox, Red Cat."* Anode = oxidation, reduction = cathode.

Mnemonic (study aid). *"Right minus left, cathode minus anode."* $E_{cell} = E_R - E_L$, with reduction potentials as printed.

4 Last 10 minutes before the exam

Top facts

1. Galvanic: chemical → electrical, spontaneous; electrolytic: electrical → chemical, non-spontaneous.
2. Electrode potential = potential at the metal–own-ion-solution interface, E.
3. Ionisation faster: metal (−) + cation layer; deposition faster: metal (+) + anion layer; Helmholtz double layer.
4. E°: 1 M, 298 K, 1 atm; single electrode potentials cannot be measured.
5. Notation: $\mathbf{Zn|Zn^{2+}(1\,M)||Cu^{2+}(1\,M)|Cu}$; anode left, | phase boundary, || salt bridge.
6. $E_{cell} = E_{cathode} - E_{anode}$ with reduction potentials (every ESA cell numerical).
7. $\Delta G = -nFE_{cell}$; $E_{cell} > 0 \Leftrightarrow$ spontaneous.
8. Daniell: 1.10 V, $-\mathbf{212.3}$ kJ mol^{-1}; Ag/Cu: 0.46 V; Fe/Au: 1.96 V (ESA, $n = 6$); Al/Cu: 2.00 V.
9. Series: Li $-\mathbf{3.03}$ ... $\mathbf{H_2}$ 0 ... Cu $+\mathbf{0.34}$, Ag $+\mathbf{0.80}$, Au $+\mathbf{1.50}$ (slide) / 1.52 (ESA).
10. Zn protects scratched iron (sacrificial); Sn does not (iron pits).

Top formulas

E4.1 · Cell potential

ESA ★★★

$$E_{cell} = E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$$

GIVES	The emf of the cell from the two electrode (reduction) potentials.
APPLIES	Every cell; with E° values it gives E°_{cell}.
UNITS	V
NOTE	Use reduction potentials for both electrodes; do not change the sign of the anode value before subtracting.

S001·9 · T001 00:15:37

E4.2 · Standard cell potential

ESA ★★★

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

GIVES	The emf when all concentrations are 1 M, gases at 1 atm, $T = 298$ K.
APPLIES	First step of every cell numerical (then Nernst, class 002).
UNITS	V
NOTE	Daniell: $0.34 - (-0.76) = 1.10$ V.

S001·9 · T001 00:15:20

E4.3 · Free energy and cell potential

ESA ★★★

$$\Delta G = -nFE_{cell} \qquad \Delta G^\circ = -nFE^\circ_{cell}$$

GIVES The free-energy change of the cell reaction (= minus the maximum electrical work).

APPLIES Deciding spontaneity; converting emf to energy.

UNITS ΔG in J mol^{-1} (E in V, F in C mol^{-1})

NOTE n = moles of electrons transferred in the balanced cell reaction; $F = 96500$ C mol^{-1}. The standard form $\Delta G^\circ = -nFE^\circ$ is stated in the class 002 derivation.

S001·9 · T001 00:16:25 · T001 00:21:24

E4.4 · Spontaneity criterion

ESA ★★★

$$E_{cell} > 0 \;\Leftrightarrow\; \Delta G < 0 \text{ (spontaneous)} \qquad E_{cell} < 0 \;\Leftrightarrow\; \Delta G > 0 \text{ (non-spontaneous)}$$

GIVES Whether the cell reaction as written proceeds on its own.

APPLIES After computing E_{cell}.

UNITS —

NOTE Galvanic cells run on reactions with $E_{cell} > 0$; electrolytic cells force reactions with $E_{cell} < 0$.

S001·9 · T001 00:16:39

E3.2 · Cell notation (Daniell cell)

ESA ★★★

$$\underbrace{\mathrm{Zn(s)\,|\,Zn^{2+}(1\,M)}}_{\text{anode (LHS)}} \;||\; \underbrace{\mathrm{Cu^{2+}(1\,M)\,|\,Cu(s)}}_{\text{cathode (RHS)}}$$

GIVES A compact representation of the cell.

APPLIES Writing any cell: anode first, salt bridge in the middle.

UNITS —

NOTE The slide writes / and // for | and ||; both are accepted.

S001·8 · T001 00:12:38

E2.1 · The two competing electrode reactions

ESA ★★

$$\underbrace{\mathrm{M \rightleftharpoons M^{n+} + \mathit{n}e^-}}_{\text{ionisation (oxidation)}} \qquad \underbrace{\mathrm{M^{n+} + \mathit{n}e^- \rightleftharpoons M}}_{\text{deposition (reduction)}}$$

GIVES The origin of the electrode potential: the faster reaction decides the sign of the charge on the metal.

APPLIES A metal M in contact with a solution of its own ions.

UNITS —

NOTE Ionisation faster: metal negative (case I). Deposition faster: metal positive (case II).

S001·5 · S001·6 · S001·7